

Keidai Iiyama

PHD CANDIDATE · STANFORD AERONAUTICS AND ASTRONAUTICS

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Education

Stanford University

Stanford, California

PHD CANDIDATE - AERONAUTICS AND ASTRONAUTICS

2021.9 - 2026.6 (Expected)

- Thesis: Design and Algorithms for Lunar Navigation Satellite Systems

The University of Tokyo

Tokyo, Japan

ME - AERONAUTICS AND ASTRONAUTICS

2019.4 - 2021.3

- Thesis: System Design and Autonomous Orbit Determination Strategy for Lunar Navigation Satellite System (Advisor: Ryu Funase)

The University of Tokyo

Tokyo, Japan

BE - AERONAUTICS AND ASTRONAUTICS

2015.4 - 2019.3

- Thesis: Navigation Satellite Constellation and Monitoring Station Arrangement for Lunar Global Navigation Satellite System (Advisor: Ryu Funase)

Awards, Fellowships, & Grants

AWARDS

- | | |
|--------|---|
| 2025.9 | ION GNSS+ 2025 Best Presentation of the Session (as first-author), for the paper (C23) |
| 2025.5 | Lunar Autonomy Challenge Top Prize, as Stanford NAV Lab team (out of 31 teams) |
| 2024.9 | ION GNSS+ 2024 Best Presentation of the Session (as first-author), for the paper (C18) |
| 2023.9 | ION GNSS+ 2023 Best Presentation of the Session (as second-author), for the paper (C13) |

FELLOWSHIPS & GRANTS

- | | |
|-----------------|--|
| 2022.1 | Student Registration Grant, 2022 ION ITM Conference |
| 2021.9 - 2022.8 | Ph.D. Student Fellowship, Stanford Aero/Astro Department |
| 2021.9 - 2026.6 | The Nakajima Foundation Doctoral Studying Abroad Fellowship, Nakajima Foundation |
| 2019.9 - 2021.8 | WINGS CFS Fellowship, WINGS CFS, The University of Tokyo |

Research Experience

Navigation and Autonomous Vehicles (NAV) Lab, Stanford University

Stanford, CA

ADVISOR: PROF. GRACE GAO

Sep, 2021 - Present

- Designed distributed time transfer algorithm from terrestrial GPS to lunar surface communication networks (C9)
- Designed Positioning and timing algorithm of lunar satellites and rovers using terrestrial GPS time-differenced carrier phase (TDCP) measurements (J9), (J2), (C11), (C12), (C14)
- Developed open-source simulator for lunar positioning, navigation, and timing (C15), (C20)
- Proposed satellite ephemeris optimization algorithms for lunar orbits (C13), (J3), (J7)
- Created distributed orbit determination and timing framework for satellite networks using covariance intersection (C17)
- Designed autonomous fault detection algorithm for satellite constellation using inter-satellite links and rigid graphs (C18), (J4)
- Analyzed trade-off space for Lunar GPS constellation design (C22), (J5)
- Proposed staged development algorithm for lunar navigation satellite constellations (C22)
- Analyzed plasmaspheric delays on GNSS signals received at lunar orbit and surface (C23), (J6)

Jet Propulsion Laboratory (332H), California Institute of Technology

Pasadena, CA

ADVISOR: DR. KAR-MING CHEUNG, DR. WILLIAM W. JUN

Jul, 2024 - Sep, 2024

- Designed Mars constellation for communication and PNT
- Proposed orbit determination and time synchronization algorithm that combines inter-satellite link and surface station link
- Related Publications: (C19)

Space Rendezvous Lab, Stanford University

ADVISOR: PROF. SIMONE D'AMICO

Stanford, CA
Sep, 2021 - May, 2022

- Created angles-only navigation algorithm for spacecraft swarms in lunar orbits
- Related Publications: (C8)

Space Systems Optimization Group, Georgia Institute of Technology (Research Collaboration)

COLLABORATOR: PROF. KOKI HO

2020

- Designed guidance, navigation, and control algorithm that combines an autoencoder, deep reinforcement learning, and a feedback controller that autonomously selects a safe landing target site and plans a divert maneuver from Lidar DEM observations during powered descent to planetary bodies.
- Related Publications: (C5), (C6)

Intelligent Space Systems Laboratory, The University of Tokyo

Bunkyo-Ku, Tokyo, Japan

ADVISOR: PROF. RYU FUNASE

Apr, 2018 - Aug, 2021

- Designed a decentralized autonomous orbit determination and clock offset estimation algorithm for lunar navigation satellites connected by crosslinks using a Schmidt-Kalman filter.
- Developed a user positioning simulator for a lunar navigation satellite system. Conducted positioning performance analysis for several constellations and lunar monitoring station arrangement patterns.
- Related Publications: (C3), (C7), (C10)

EQUULEUS Project Team

ADVISOR: PROF. RYU FUNASE

Oct, 2017 - Aug, 2021

- EQUULEUS is a 6U CubeSat lunar mission jointly proposed by JAXA and the University of Tokyo. It was selected as the secondary payload of the EM-1 Mission and achieved a successful lunar flyby.
- Designed and implemented flight software to control heaters and detect faults of the thermal sub-system.
- Related Publications: (J1), (C1), (C2), (C4)

Publications

JOURNAL PAPERS UNDER REVIEW

(J9) **Iiyama, K.**, and Gao, G., "GNSS-based Lunar Orbit and Clock Estimation With Stochastic Cloning UD Filter", Submitted to Journal of Guidance, Control, and Dynamics, Under Review, <https://doi.org/10.48550/arXiv.2601.16393>

(J8) Dai, A., Vila, C.G., Wu, A., **Iiyama, K.**, Coimbra, K., Deng, T., and Gao, G., "Full Stack Navigation, Mapping, and Planning for the Lunar Autonomy Challenge", Submitted to NAVIGATION: Journal of the Institute of Navigation, Under Review

(J7) **Iiyama, K.**, and Gao, G., "Ephemeris and Almanac Design for Lunar Navigation Satellites", Submitted to IEEE Transactions on Aerospace and Electronic Systems, Under Review, <https://doi.org/10.48550/arXiv.2510.25161>

(J6) **Iiyama, K.**, and Gao, G., "Ionospheric and Plasmaspheric Delay Characterization for Lunar Terrestrial GNSS Receivers with Global Core Plasma Model", Submitted to NAVIGATION: Journal of the Institute of Navigation, Minor revision, <https://doi.org/10.48550/arXiv.2510.10059>

(J5) **Iiyama, K.**, and Gao, G., "Trade-off Analysis for Lunar Augmented Navigation Service (LANS) Constellation Design", Submitted to NAVIGATION: Journal of the Institute of Navigation, Under Review, <https://doi.org/10.48550/arXiv.2510.16030>

JOURNAL PAPERS

(J4) **Iiyama, K.**, Neamati, D., and Gao, G., "Satellite Autonomous Clock Fault Monitoring with Inter-Satellite Ranges Using Euclidean Distance Matrices", NAVIGATION: Journal of the Institute of Navigation, Accepted, <https://doi.org/10.48550/arXiv.2505.03820>

(J3) Cortinovis, M., **Iiyama, K.**, and Gao, G., "Satellite Ephemeris Parameterization Methods to Support Lunar Positioning, Navigation, and Timing Services", NAVIGATION: Journal of the Institute of Navigation, Vol. 71, Issue 4, 2024, <https://doi.org/10.33012/navi.664>

(J2) **Iiyama, K.**, Bhamidipati, S., and Gao, G., "Precise Positioning and Timekeeping in Lunar Orbit via Terrestrial GPS Time-Differenced Carrier-Phase Measurements", NAVIGATION: Journal of the Institute of Navigation, Vol. 71, Issue 1, 2024
<https://doi.org/10.33012/navi.635>

(J1) Funase, R., Ikari, S., Miyoshi, K., Kawabata, Y., Nakajima, S., Nomura, S., Funabiki, N., Ishikawa, A., Kakiyama, K., Matsushita, S., Takahashi, R., Yanagida, K., Mori, D., Murata, Y., Shibukawa, T., Suzumoto, R., Fujiwara, M., Tomita, K., Aohama, H., **Iiyama, K.**, et al., "Mission to Earth-Moon Lagrange Point by a 6U CubeSat: EQUULEUS", IEEE Aerospace & Electro. Systems Magazine, Vol. 35, No.3, pp.30-44, 2020, <https://doi.org/10.1109/MAES.2019.2955577>

CONFERENCE PROCEEDINGS *: Equal Contribution

(C25) Matthias Kura, **Iiyama, K.**, and Gao, G., “Sidelobe vs. Side of Main Lobe: LuGRE Experiment Data Analysis and Filtering”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2026), Orlando, FL, 2026, Abstract Submitted

(C24) Matthias Kura, **Iiyama, K.**, and Gao, G., “Optimal Reconfiguration of Lunar Navigation Constellations”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2026), Orlando, FL, 2026, Abstract Submitted

(C23) **Iiyama, K.**, and Gao, G., “Ionospheric and Plasmaspheric Delay Characterization and Mitigation Methodologies for Lunar Terrestrial GNSS Receivers”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2025), Baltimore, MD, 2025, **Best Presentation of the Session Award**

(C22) **Iiyama, K.**, and Gao, G., “Constellation Design and Staged Development for the Lunar Navigation Satellite System”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2025), Baltimore, MD, 2025

(C21) Dai, A., Wu, A., **Iiyama, K.**, Vila, G.C., Coimbra, K., Deng, T., and Gao, G., “Full Stack Navigation, Mapping, and Planning for the Lunar Autonomy Challenge”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2025), Baltimore, MD, 2025

(C20) *Vila, G.C., ***Iiyama, K.**, and Gao, G., “LuPNT: An Open-Source Simulator for Lunar Communications, Positioning, Navigation, and Timing”, 2025 IEEE Aerospace Conference, Big Sky, MT, 2025

(C19) **Iiyama, K.**, Jun, W.W., Bhamidipati, S., Gao, G., Cheung, K.-M., “Orbit Determination and Time Synchronization for the Future Mars Relay and Navigation Constellation”, 2025 IEEE Aerospace Conference, Big Sky, MT, 2025

(C18) **Iiyama, K.**, Neamati, D., and Gao, G., “Autonomous Constellation Fault Monitoring with Inter-satellite Links: A Rigidity-Based Approach”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2024), Baltimore, MD, 2024, **Best Presentation of the Session Award**

(C17) ***Iiyama, K.**, *Vila, G.C., and Gao, G., “Contact Plan Optimization and Distributed State Estimation for Delay Tolerant Satellite Networks”, 2024 IEEE Aerospace Conference, Big Sky, MT, 2024

(C16) *Shimane, Y., and ***Iiyama, K.**, “Methods for Dual-Objective High Energy Tour Design”, 2023 AAS/AIAA Astrodynamics Specialist Conference, Big Sky, MT, 2023

(C15) ***Iiyama, K.**, *Vila, G.C., and Gao, G., “LuPNT: Open-Source Simulator for Lunar Positioning, Navigation, and Timing”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2023), Denver, CO, 2023

(C14) **Iiyama, K.**, and Gao, G., “Positioning and Timing of Distributed Lunar Satellites via Terrestrial GPS Differential Carrier Phase Measurements”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2023), Denver, CO, 2023

(C13) Cortinovis, M., **Iiyama, K.**, and Gao, G., “Satellite Ephemeris Approximation Methods to Support Lunar Positioning, Navigation, and Timing Services”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2023), Denver, CO, 2023, **Best Presentation of the Session Award**

(C12) **Iiyama, K.**, Bhamidipati, S., and Gao, G., “Terrestrial GPS Time-Differenced Carrier-Phase Positioning of Lunar Surface Users”, 2023 IEEE Aerospace Conference, Big Sky, MT, 2023

(C11) **Iiyama, K.**, Bhamidipati, S., and Gao, G., “Precise Positioning and Timekeeping in Lunar Orbit via Terrestrial GPS Time-Differenced Carrier-Phase Measurements”, Proceedings of the 2023 International Technical Meeting of The Institute of Navigation, Long Beach, CA, 2023

(C10) Murata, M., Koga, M., Nakajima, Y., Yasumitsu, R., Araki, T., Makino, K., Akiyama, K., Yamamoto, T., Tanabe, K., Kogure, S., Sato, N., Toyama, D., Kitamura, K., Miyasaka, K., Kawaguchi, T., Sato, Y., Kakihara, K., Shibukawa, T., **Iiyama, K.**, Tanaka, T., “Lunar navigation satellite system: Mission, system overview, and demonstration”, 39th International Communications Satellite Systems Conference (ICSSC 2022), Stresa, Italy, 2022

(C9) Bhamidipati, S., ***Iiyama, K.**, *Mina, T., and Gao, G., “Time-Transfer from Terrestrial GPS for Distributed Lunar Surface Communication Networks”, 2022 IEEE Aerospace Conference, Big Sky, MT, 2022

(C8) **Iiyama, K.**, Kruger, J., and D’Amico, S., “Autonomous Distributed Angles-Only Navigation and Timekeeping in Lunar Orbit”, Proceedings of the 2023 International Technical Meeting of The Institute of Navigation, Long Beach, CA, 2022, **Student Registration Grant**

(C7) **Iiyama, K.** and Funase, R., “Autonomous and Decentralized Orbit Determination and Clock Offset Estimation of Lunar Navigation Satellites Using GPS Signals and Inter-satellite Ranging”, Proceedings of the Institute of Navigation GNSS+ conference (ION GNSS+ 2021), St.Louis, MO, USA, 2021

(C6) Tomita, K., Skinner, K., **Iiyama, K.**, Jagatia, B.A., Nakagawa, T., and Ho, K., “Real-Time Terrain Mapping and Processing for Safe Landing via Deep Neural Networks”, ASCEND, Las Vegas, NV, 2020

(C5) **Iiyama, K.**, Tomita, K., Jagatia, B.A., Nakagawa, T., and Ho, K., “Deep Reinforcement Learning for Safe Landing Site Selection with Concurrent Consideration of Divert Maneuver”, 2020 AAS/AIAA Astrodynamics Specialist Conference, Online, 2020

(C4) Shibukawa, T., Matsushita, S., **Iiyama, K.**, Ishikawa, A., Nishii, K., and Funase, R. “Flight Model Thermal Design and Validation for a 6U Deep Space Cubesat EQUULEUS”, 50th International Conference on Environmental Systems, Lisbon, Portugal, 2020 (conference postponed)

(C3) **Iiyama, K.**, “Optimization of the Navigation Satellite Constellation and Lunar Monitoring Station for Lunar Global Navigation Satellite System”, 32nd International Symposium on Space Technology and Science, Fukui, Japan, 2019

(C2) Matsushita, S., Shibukawa, T., **Iiyama, K.**, and Funase, R., “Thermal Design and Validation for a 6U Cubesat EQUULEUS under Constraints Tightly Coupled with Orbital Design and Water Propulsion System”, 49th International Conference on Environmental Systems, Boston, MA, The United States, 2019

(C1) Shibukawa, T., Matsushita, S., **Iiyama, K.**, and Funase, R., “Reflection and Verification of Thermal Design under Tightly-Coupled Constraints to the 6U Deep Space CubeSat EQUULEUS”, 32nd International Symposium on Space Technology and Science, Fukui, Japan, 2019

MAGAZINE ARTICLES

(M1) **Iiyama, K.**, Pullen, S., and Gao, G., “Can Satellite-based Radionavigation be Extended to the Moon and Other Extraterrestrial Bodies?”, Inside GNSS, Vol. 20, Issue 5, 2025

OTHER INVITED TALKS

(T4) “Fundamentals of Lunar PNT: Reference Frames, Time Systems, and Cislunar Dynamics”, ION GNSS+ Tutorial Session, September, 2026 (Scheduled)

(T3) “Designing a GPS System for the Moon”, Stanford planetary science and exploration seminar, May, 2026 (Scheduled)

(T2) “Satellite Ephemeris Parameterization Methods to Support Lunar PNT Services”, ION Webinar, November, 2024

(T1) “Precise and Robust On-board Positioning, Navigation, and Timing in the Lunar Regime”, NASA Goddard Space Flight Center, April, 2024

Teaching Experience

Spring 2026 (Scheduled)	AA278: Lunar Positioning, Navigation, and Timing , scheduled, teaching with Prof. Grace Gao, Graduate Course Assistant, Stanford University
Fall 2024, 2025	AA272: Global Positioning Systems , taught by Prof. Grace Gao, Graduate Course Assistant, Stanford University
Fall 2019	Orbital Mechanics , taught by Prof. Ryu Funase, Graduate Teaching Assistant, The University of Tokyo

Mentoring

2025.9 -	Matthias Kura , Ph.D. Student, Stanford University
2025.9 -	Vivian Sattler , Ph.D. Student, Stanford University
2025.1 - 2025.3	Pauline de la Hougue Moran , M.S. Student, Stanford University
2024.9 -	Kaila Coimbra , Ph.D. Student, Stanford University
2023.1 -	Guillem Casadesus Vila , Ph.D. Candidate, Stanford University
2023.1 - 2024.9	Marta Cortinovis , Ph.D. Candidate, Stanford University
2022.9 - 2023.3	Guillem Rueda Oller , Ph.D. Student, Stanford University

Services

PEER REVIEW

IEEE Transactions on Aerospace and Electronic Systems

NAVIGATION: Journal of the Institute of Navigation

CONFERENCE ORGANIZATION

ION GNSS+ 2026 Session Chair for “*Cislunar, Lunar and Martian Positioning, Navigation, and Timing*” (Scheduled)

STUDENT ORGANIZATION

Served as a financial officer for Engineering Students for Diversity, Equity, and Inclusion (ES4DEI). Also hosted a sharing stories event, with the aim to increase empathy by sharing first-hand experiences and listening to others' experiences